Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

AI-forward software engineer with 2+ years building **EdTech products, LLM workflows, React/Python applications, and PostgreSQL systems**; ships features end-to-end using Cursor, custom orchestration, evaluation, regression testing, RAG, and cloud deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **student-data and academic workflow services** across Python, FastAPI, PostgreSQL, SQL Server, and cloud storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Owned fault-tolerant LLM orchestration processing 1,000+ AI jobs with structured states, failure isolation, validation, and regression checks, improving throughput 3x while sustaining 99%+ reliability across 5,000+ monthly workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Shipped a **full-stack AI grading product** with React dashboards, Flask APIs, student records, rubric generation, feedback, and multithreaded processing, serving 150+ instructors and 10,000+ daily requests.
- Integrated production OpenAI, Claude, and Qwen through custom prompt chains and structured outputs, evaluating results against human grading to achieve 90%+ consistency while reducing feedback turnaround 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **scalable cross-platform Python/Java services and custom automation** using REST APIs, AWS Glue, S3, SQL, and reusable design patterns, improving response times 30% and pipeline efficiency 68%.
- Maintained 20+ unit and integration tests with PyTest and Mockito while collaborating with Microsoft and GitHub engineers through requirements, code reviews, Git, Jenkins, CI/CD, and production debugging.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Shipped **scalable React and Node.js product features** across eight modular services for user records, permissions, dashboards, notifications, and real-time updates, synchronizing 2M+ records with secure role-based access.
- Owned features through implementation, automated testing, deployment, and direct monitoring with Docker, Jenkins, Sentry, Azure Monitor, and Kibana, achieving 95%+ build validation while reducing deployment time 50%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **machine-learning data and evaluation pipelines** over 100,000+ trajectories, translating ambiguous research goals into scoped experiments, quantitative metrics, and production-quality Python workflows while improving planning performance 18%.
- Designed repeatable model evaluations across 10,000 driving episodes, analyzing failures, long-horizon behavior, and closed-loop stability to improve performance 15% while documenting trade-offs and iterating directly with researchers.

Skills

Full Stack: TypeScript, React.js, Next.js, Node.js, Python, FastAPI, Flask, PostgreSQL, REST APIs
AI Engineering: OpenAI, Anthropic Claude, Qwen, LLM orchestration, prompt chains, structured outputs, tool use
Retrieval: RAG, FAISS, Pinecone, vector search, embeddings, semantic retrieval, incremental indexing
Evaluation: Custom evals, regression testing, human comparison, failure analysis, model metrics, PyTest, Jest
Workflow: Cursor, AI-assisted development, Git, GitHub Actions, AWS, GCP, Docker, CI/CD, end-to-end ownership

Projects

Big-Brain Live | *RAG, FAISS, PostgreSQL, LLMs*

- Shipped an **LLM-powered RAG application** processing 50,000+ documents with authenticated uploads, PostgreSQL metadata, incremental indexing, FAISS vector retrieval, summarization, and collaborative workflows, reducing retrieval latency 40% and review time 60%.

CaptionCraft Live | *AI Evals, Python, APIs*

- Built a **custom model-evaluation benchmark** across four vision architectures, 10,000 images, and 100,000 captions, analyzing regression results and failures to improve accuracy from 81% to 96% while shipping inference APIs.

Chat2SQL Code | *TypeScript, Java, AI Tooling*

- Extended an **open-source AI development tool** across TypeScript and Java modules, supporting 10+ databases with schema-aware retrieval, structured query generation, Docker, Maven, Node.js, validation, and cross-platform workflows.